

NEUROSENSE

QUANTITATIVE TRANSLATIONAL EEG INTELLIGENCE

Comprehensive Brain Health Analysis Report



Patient Name	Test Patient
Date of Recording	2025-11-28
Age	35 years
Gender	Male
Handedness	Right

*This report provides quantitative EEG analysis for brain health assessment.
Generated by NeuroSense Intelligence™*

Introduction

Important Notice

This report is NOT a medical diagnosis. It provides quantitative analysis of brain wave activity recorded through electroencephalography (EEG). The information presented should be interpreted by qualified healthcare professionals in conjunction with clinical assessment and other diagnostic information. This analysis is for informational and research purposes only.

Patient-Reported Symptoms

Please indicate which symptoms you are currently experiencing:

- | | | |
|---|---|--|
| ☐ ADHD | ☐ Depression | ☐ Anxiety |
| ☐ OCD | ☐ Autism | ☐ Schizophrenia |
| ☐ Memory Issues | ☐ Insomnia | ☐ Epilepsy |
| ☐ Substance Use Disorder | ☐ Traumatic Brain Injury | ☐ Tinnitus |
| ☐ Dyslexia | | |

About EEG Recording

Electroencephalography (EEG) measures electrical activity in the brain using electrodes placed on the scalp. This analysis used the International 10-20 system, a standardized electrode placement method that ensures consistent and reliable measurements across different recording sessions.

Recording performed using 19-channel EEG with the International 10-20 electrode system.

Electrode Locations: Fp1, Fp2 (frontal poles), F7, F3, Fz, F4, F8 (frontal), T3, C3, Cz, C4, T4 (central), T5, P3, Pz, P4, T6 (parietal), O1, O2 (occipital)

Sample EEG waveforms represent brain electrical activity across different regions, recorded simultaneously from all 19 electrode sites.

Understanding Brain Waves

Your brain produces electrical signals at different frequencies, known as brain waves. Each frequency band is associated with different mental states and cognitive functions. Understanding your brain wave patterns helps identify areas of strength and opportunities for improvement.

Delta (1-3 Hz)

Deep sleep, restoration, healing

Theta (4-8 Hz)

Relaxation, creativity, meditation

Alpha (8-12 Hz)

Calm, focused state, relaxed awareness

SMR (12-15 Hz)

Sensorimotor rhythm, focused relaxation

Beta (15-30 Hz)

Active thinking, alertness, concentration

A healthy brain shows appropriate levels of each frequency band depending on the state of consciousness and task being performed. The balance and distribution of these waves provide insights into cognitive function, emotional regulation, and overall brain health.

Numbers at a Glance

This section provides a quick overview of your key QEEG metrics measured under two conditions: Eyes Closed (resting state) and Eyes Open (active state). These measurements help assess different aspects of brain function and regulation.

Eyes Closed Condition (Resting State)

Metric	Value	Status
Alpha Peak Frequency	10.3 Hz	Normal
Alpha/Beta Ratio (Pz)	N/A	Low
Frontal Alpha Asymmetry	N/A	Right Dominant
Posterior Dominant Rhythm	10.3 Hz	Normal

Eyes Open Condition (Active State)

Metric	Value	Status
Theta/Beta Ratio (Focus)	NaN	Elevated
Arousal Score (HiBeta/Beta)	0.78	Normal
Frontal Theta Activity	3.8 ;Åk	Normal
Beta Activity (Alertness)	3.2 ;Åk	Normal

Frontal Alpha Asymmetry

Measures the balance of alpha activity between left and right frontal regions. Imbalances can indicate emotional regulation patterns.

Detailed analysis for this metric is not available in the current dataset.

Total Brain Coherence

Measures how well different brain regions communicate and synchronize. Higher coherence indicates better brain network integration.

Detailed analysis for this metric is not available in the current dataset.

Alpha Peak Frequency

The dominant alpha frequency indicates brain processing speed and cognitive efficiency. Higher is generally better.

Detailed analysis for this metric is not available in the current dataset.

Posterior Dominant Rhythm (PDR)

The predominant rhythm in the back of the brain during rest. PDR should match age-appropriate norms.

Detailed analysis for this metric is not available in the current dataset.

Focus & Attention Score

Based on theta/beta ratio. Lower ratios indicate better attention regulation and focus capability.

Detailed analysis for this metric is not available in the current dataset.

Arousal Score

Measures high beta relative to beta activity. Indicates stress, anxiety, or hyperarousal levels.

Detailed analysis for this metric is not available in the current dataset.

Relaxation Score

Alpha/beta ratio measuring ability to relax and disengage from active processing.

Detailed analysis for this metric is not available in the current dataset.

Peak Performance Ratio

Gamma/alpha ratio indicating cognitive processing capacity and mental performance.

Detailed analysis for this metric is not available in the current dataset.

Your Brain Type

Type 1

Relaxed and Balanced Brain

Overall Score

12/21

Your brain shows healthy alpha activity and balanced regulation. You likely experience good mood stability, calm focus, and effective stress management.

Key Characteristics

- High alpha peak frequency (> 10 Hz)
- Good alpha/beta ratio (relaxation capability)
- Low arousal score (< 1)
- Balanced frontal asymmetry
- Good theta/beta ratio (< 1.5)

Your Strengths

- ' Excellent stress management
- ' Calm and focused mental state
- ' Good emotional regulation
- ' Healthy sleep patterns

Personalized Recommendations

Based on your brain type assessment, here are personalized recommendations to optimize your brain health and cognitive function. These suggestions are designed to support your unique brain patterns and promote overall wellness.

Nootropic Suggestions

Rhodiola Rosea

L-Theanine

Ashwagandha

Supplement Support

Omega-3 Fatty Acids

Vitamin B Complex

Magnesium

Breathing Exercises

Diaphragmatic Breathing

Box Breathing

Meditation Practices

Mindfulness Meditation

Body Scan

Physical Exercises

Yoga

Walking

Swimming

Important: These recommendations are for informational purposes only. Please consult with qualified healthcare professionals before starting any new supplement, nootropic, or exercise regimen. Individual responses may vary.

Appendix: Breathing Techniques

Breathing exercises are powerful tools for regulating brain activity and improving mental health. Each technique targets specific brain wave patterns and can help optimize your cognitive function, emotional regulation, and stress management. Practice these techniques regularly for best results.

Alternate Nasal Breathing (Nadi Shodhana)

Evidence-Based Benefits:

- Balances left and right brain hemispheres
- Improves frontal lobe coherence
- Reduces anxiety and promotes calmness
- Enhances focus and mental clarity

How to Practice:

1. Sit comfortably with a straight spine
2. Use your right thumb to close your right nostril
3. Inhale slowly through your left nostril (4 counts)
4. Close left nostril with ring finger, release right nostril
5. Exhale through right nostril (4 counts)
6. Inhale through right nostril (4 counts)
7. Close right nostril, release left nostril
8. Exhale through left nostril (4 counts)
9. This completes one round. Repeat for 5-10 minutes

4-7-8 Breathing

Evidence-Based Benefits:

- Activates parasympathetic nervous system
- Reduces high beta activity (stress)
- Improves alpha/beta ratio
- Promotes rapid relaxation and better sleep

How to Practice:

1. Sit or lie down in a comfortable position
2. Place tongue tip behind upper front teeth
3. Exhale completely through mouth with a whoosh sound
4. Close mouth and inhale quietly through nose for 4 counts
5. Hold breath for 7 counts
6. Exhale completely through mouth for 8 counts
7. Repeat this cycle 4 times
8. Practice twice daily for best results

Humming Bee Breath (Bhramari)

Evidence-Based Benefits:

- Increases alpha and theta coherence
- Calms racing thoughts and anxiety
- Improves concentration
- Stimulates vagus nerve for relaxation

How to Practice:

1. Sit comfortably with eyes closed
2. Place index fingers on ears or temples
3. Inhale deeply through nose
4. Exhale while making a humming sound like a bee
5. Feel the vibration in your head
6. Continue for 5-10 breaths
7. Notice the calming effect after each round

Diaphragmatic Breathing

Evidence-Based Benefits:

- Increases alpha activity and relaxation
- Reduces arousal score (HiBeta/Beta ratio)
- Improves oxygen flow to brain
- Strengthens breathing efficiency

How to Practice:

1. Lie down or sit with back supported
2. Place one hand on chest, one on belly
3. Breathe in slowly through nose (4-5 counts)
4. Feel your belly rise while chest stays still
5. Exhale slowly through pursed lips (6-7 counts)
6. Belly falls inward
7. Practice for 5-10 minutes daily

Cyclic Sighing

Evidence-Based Benefits:

- Rapidly reduces stress and anxiety
- Lowers high beta activity
- Improves emotional regulation
- Scientifically proven to calm nervous system

How to Practice:

1. Inhale deeply through nose until lungs feel full
2. Take a second, shorter inhale to maximally fill lungs
3. Exhale very slowly through mouth
4. Make the exhale longer than the inhale
5. Repeat for 5 minutes
6. Can be done anytime you feel stressed

Bhastrika (Breath of Fire)

Evidence-Based Benefits:

- Increases beta activity and alertness
- Energizes the brain
- Improves focus and concentration
- Clears mental fog

How to Practice:

1. Sit in comfortable position with straight spine
2. Take a deep breath in through nose
3. Forcefully exhale through nose
4. Immediately forcefully inhale through nose
5. Continue rapid, forceful breathing for 10-20 breaths
6. End with a deep inhale and slow exhale
7. Rest and observe the energizing effects
8. Caution: Avoid if you have high blood pressure

Wim Hof Method

Evidence-Based Benefits:

- Increases stress resilience
- Improves focus and mental clarity
- Enhances mood and energy
- Strengthens immune system

How to Practice:

1. Sit or lie down comfortably
2. Take 30-40 deep breaths (full inhale, passive exhale)
3. After last exhale, hold breath as long as comfortable
4. When you need to breathe, inhale deeply
5. Hold breath for 15 seconds
6. Exhale and return to normal breathing
7. Repeat cycle 3 times
8. Practice in safe environment only

Box Breathing (Square Breathing)

Evidence-Based Benefits:

- Balances nervous system
- Improves focus and attention
- Reduces theta/beta ratio
- Promotes calm alertness

How to Practice:

1. Sit comfortably with straight posture
2. Exhale all air from lungs
3. Inhale through nose for 4 counts
4. Hold breath for 4 counts
5. Exhale through mouth for 4 counts
6. Hold empty lungs for 4 counts
7. Repeat for 5-10 minutes
8. Used by Navy SEALs for stress management

Rapid Cycling Breath

Evidence-Based Benefits:

- Quickly energizes the brain
- Increases alertness and beta activity
- Breaks mental fatigue
- Improves cognitive performance

How to Practice:

1. Stand or sit with good posture
2. Breathe in and out rapidly through nose
3. Keep breaths shallow and quick
4. Maintain rhythm for 30 seconds
5. Rest for 30 seconds with normal breathing
6. Repeat 3-5 times
7. Best used when feeling sluggish or unfocused

Getting Started

Choose 2-3 techniques that resonate with your brain type and current needs. Start with 5 minutes daily and gradually increase duration. Consistency is more important than duration. Track your progress and notice how different techniques affect your mental state and cognitive performance.

Safety Note:

If you experience dizziness, discomfort, or unusual sensations during any breathing exercise, stop immediately and return to normal breathing. Consult a healthcare professional if symptoms persist. Some techniques may not be suitable for individuals with certain medical conditions.

